PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Protected entry still missing; OV chopper and fault memory populated on dedicated pages.

GEN DRV8353S driver and three low-side current channels

File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only

File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels

File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only

File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries

File: external-boundaries.kicad_sch

Bus-fed auxiliary and independent safety supply

File: aux-supply.kicad_sch

Autonomous OV chopper comparator and rail / brake monitors

File: protection-comparators.kicad_sch

Release-before-rearm DFF memory

File: fault-latch.kicad_sch

LVC Schmitt fault inputs without series diode offsets

File: fault-interfaces.kicad_sch

Independent brake chopper power stage

File: brake-chopper.kicad_sch

STM32G474VET6 LQFP100: power, reset, debug and control

File: mcu-core.kicad_sch

MCU 3V3 decoupling — external supervised rail

File: mcu-power.kicad_sch

Six CSA ADC input filters and brake status sense

File: mcu-analog.kicad_sch

3V3 Hall interfaces and optional quadrature encoder

File: mcu-sensors.kicad_sch

GEN independent raw CSA window trips

File: gen-fast-oc.kicad_sch

HUB independent raw CSA window trips

File: hub-fast-oc.kicad_sch

Ratiometric OC window and wired sink fault

File: fast-oc-reference.kicad_sch

Bounded bench DC entry — active pairs inhibited

File: dc-entry.kicad_sch

LEFT supervised binary brake interface

File: brake-intent-left.kicad_sch

RIGHT supervised binary brake interface

File: brake-intent-right.kicad_sch

Rider dump PWM OR independent emergency overvoltage

File: brake-dump-command.kicad_sch

OFFBOARD supervised switch end-of-line network — candidate

File: brake-switch-harness.kicad_sch

LEFT selected FSR low-voltage bench frontend

File: fsr-left.kicad_sch

LEFT FSR candidate ESD and secondary clamps — unqualified

File: fsr-protection-left.kicad_sch

RIGHT selected FSR low-voltage bench frontend

File: fsr-right.kicad_sch

RIGHT FSR candidate ESD and secondary clamps — unqualified

File: fsr-protection-right.kicad_sch

MAIN back-to-back candidate — hardware inhibited

File: entry-main.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheets: /
File: powerboard-reva.kicad_sch

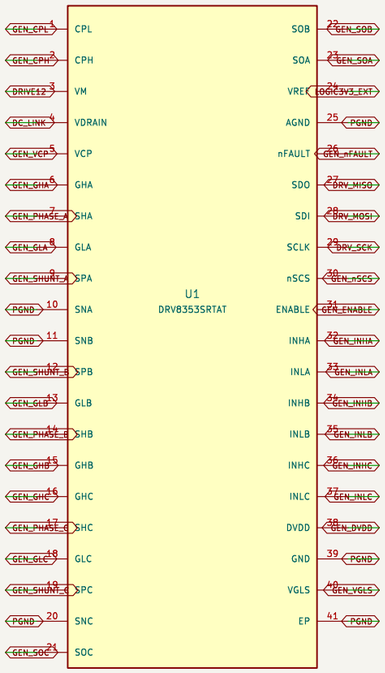
Title: Pedal-by-wire: hierarchical engineering review

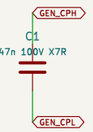
Size: A1	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 1/32

TI SLVSDY6A pp6–7 pin table. 40–pin RTA + exposed pad. SPI variant, NOT hardware variant.

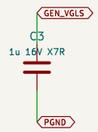
VM supplied by bus–fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

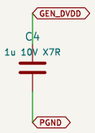
RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.

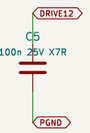


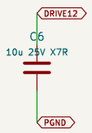




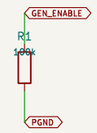


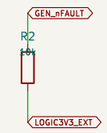


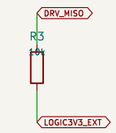


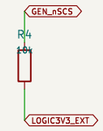












ENABLE pull–down defaults off; not a safety–rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

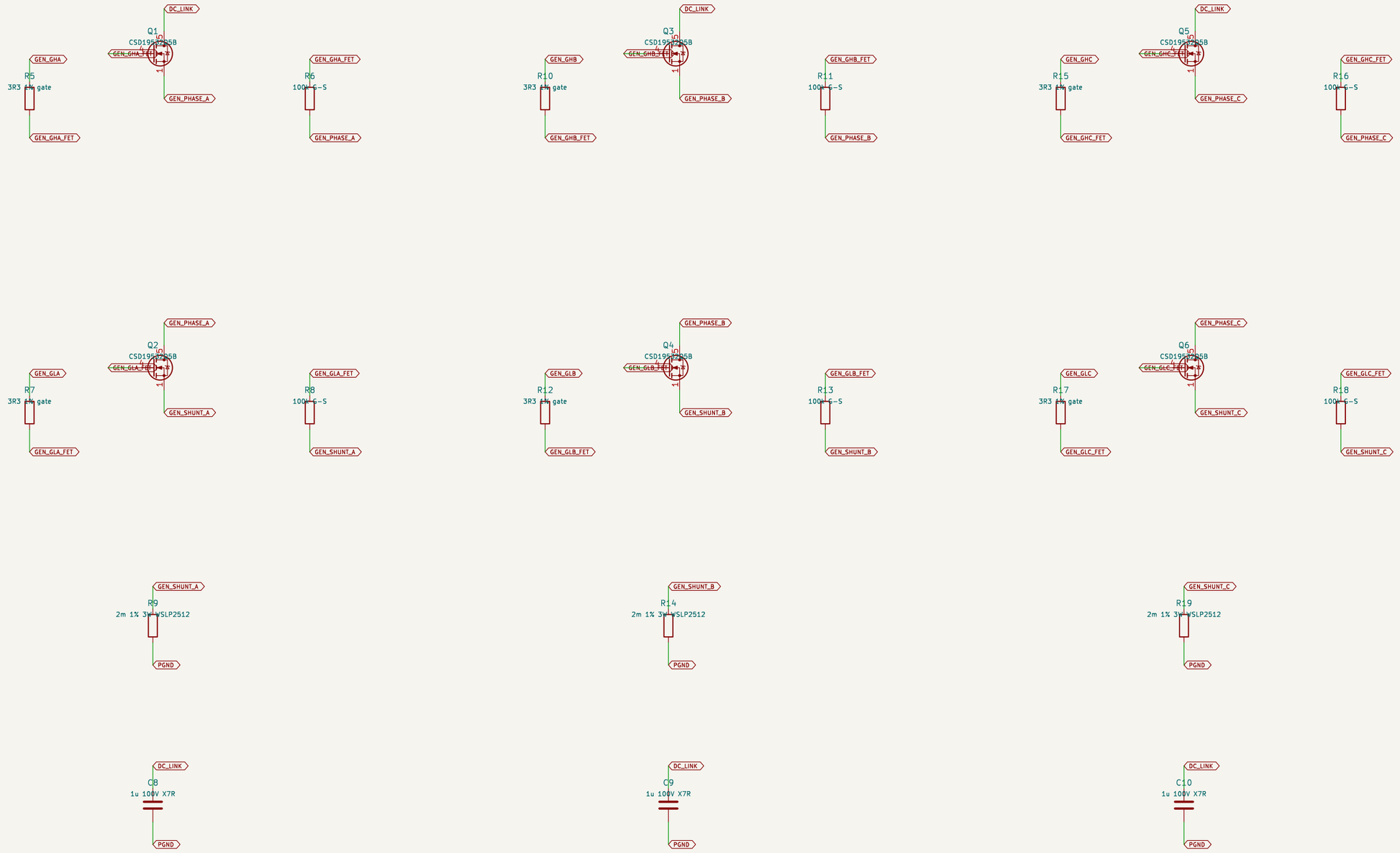
Sheet: /GEN DRV8353S driver and three low–side current channels/
File: gen–driver.kicad_sch

Title: GEN DRV8353S driver and three low–side current channels

Size: A3	Date:	**Rev: A–PARTIAL**
KiCad E.D.A. 9.0.9		Id: 2/32

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=3R3 (RAIL−10, sized in size_gate_resistors.py). Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

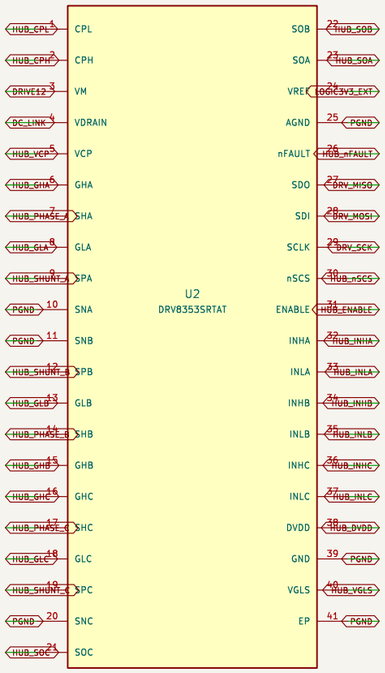
Title: GEN 100 V MOSFET bridge — schematic only

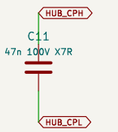
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 3/32

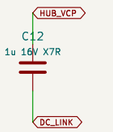
TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

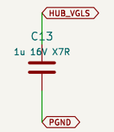
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

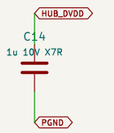
RELEASE BLOCKERS: firmware, OC timing/coverage and transient/thermal qualification unfinished.





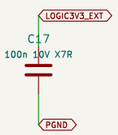


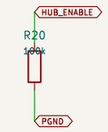


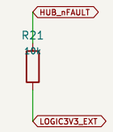


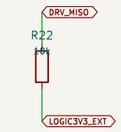


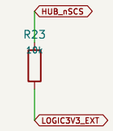












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

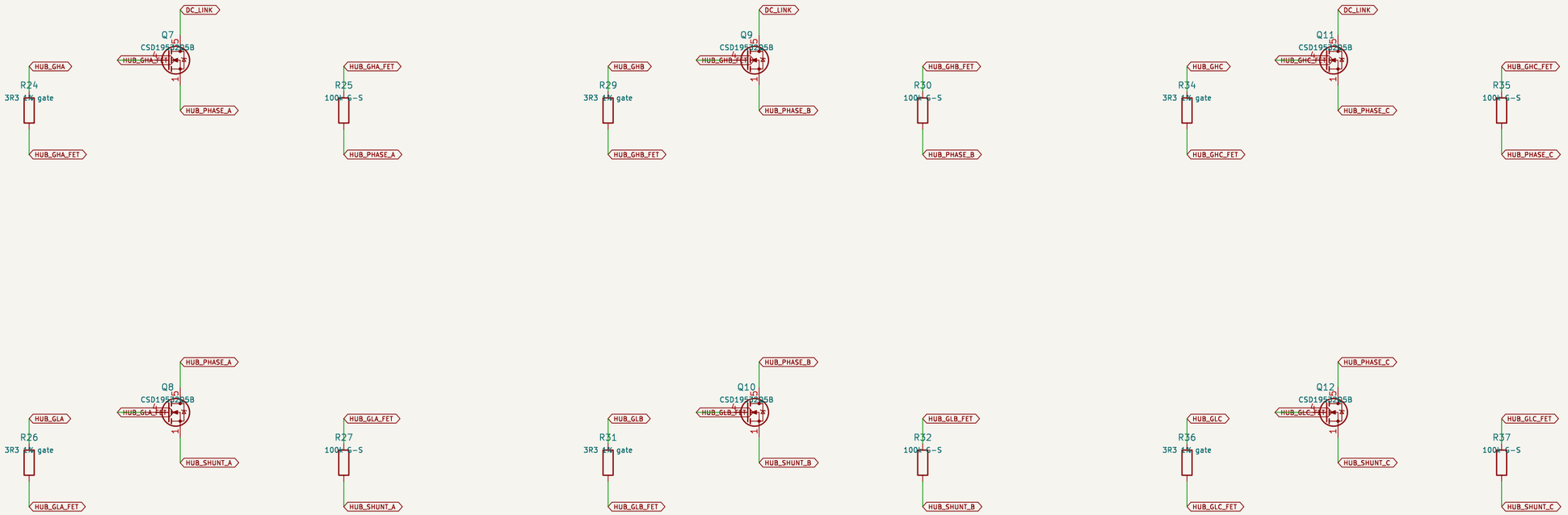
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

Title: HUB DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 4/32

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=3R3 (RAIL−10, sized in size_gate_resistors.py). Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

Title: HUB 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 5/32

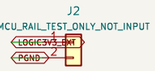
OFFBOARD interfaces: 3V3 supply, brake loop and protected bus entry.

DO NOT ENERGIZE. Protected entry unfinished; MCU and protection remain unqualified.

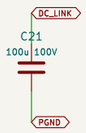












Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Populated: OV chopper, fault latch, brake loop, watchdog, aux supply. Added: raw CSA window trip; timing/coverage unqualified.

MCU and sensor interfaces populated; firmware, thermal sensors, storage/BMS acceptance missing.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

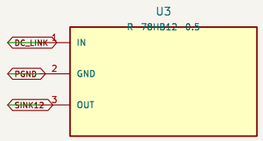
Title: Unfinished safety / control / supply boundaries

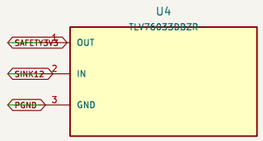
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 6/32

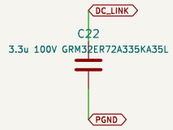
#FLG009
PWR_FLAG

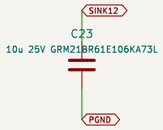
LOGIC3V3_EXT: MCU 3V3 rail brought in at the boundary connector J2

R-78HB12-0.5: 17–72V operating, 75V absolute. Entry transient protection still REQUIRED.



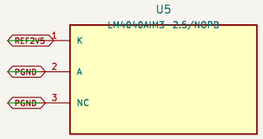




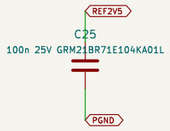












Safety supply is NOT the MCU supply. LOGIC3V3_EXT remains a supervised, external MCU rail.

Supply total/current/thermal budget provisional; no galvanic isolation or stored-energy ride-through.

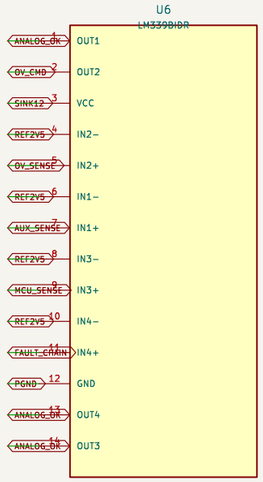
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

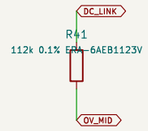
Sheet: /Bus-fed auxiliary and independent safety supply/
File: aux-supply.kicad_sch

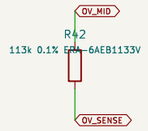
Title: Bus-fed auxiliary and independent safety supply

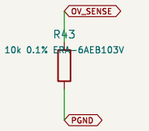
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 7/32

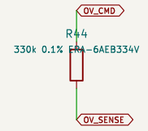
Comparator powered from SINK12; open collectors pulled to SAFETY3V3. No MCU in OV path.



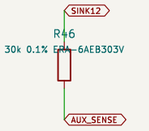


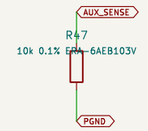


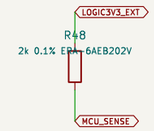


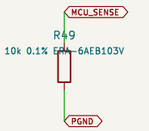


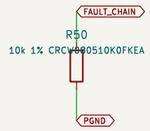




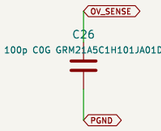














FAULT_CHAIN must close to SAFETY3V3 through normally-closed E-stop + dump thermal switches (NOT rider levers). Open wire trips.

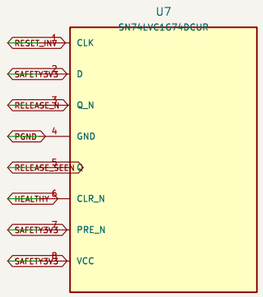
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

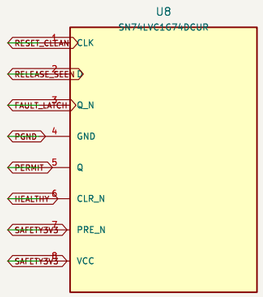
Sheet: /Autonomous OV chopper comparator and rail / brake monitors/
File: protection-comparators.kicad_sch

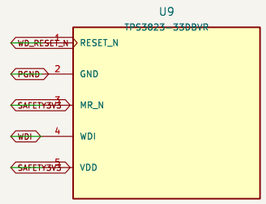
Title: Autonomous OV chopper comparator and rail / brake monitors

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 8/32

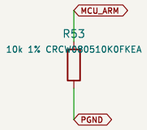
UNSAFE DRAFT: healthy release edge THEN press edge required. No automatic reset.













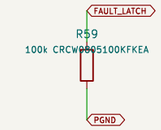


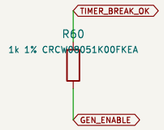


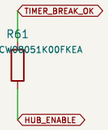






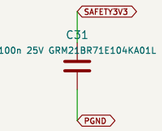












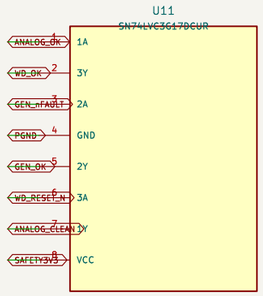
TPS3823 non-A: falling WDI only; hold low during RESET. Stuck RESET stays disabled.

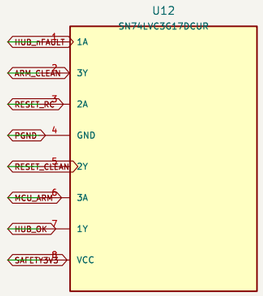
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

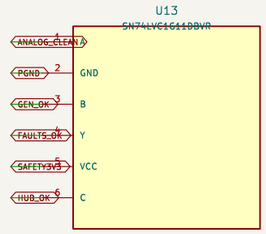
Sheet: /Release-before-rearm DFF memory/
File: fault-latch.kicad_sch

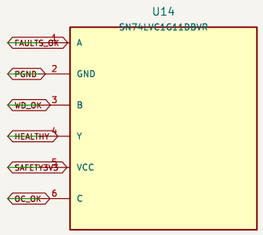
Title: Release-before-rearm DFF memory

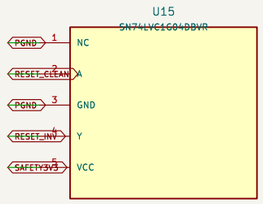
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 9/32

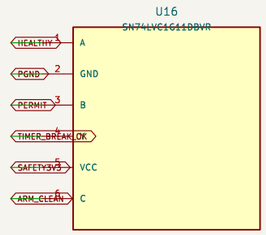
























Default OFF via async supervisor clear. Analog ramp/metastability NOT validated.

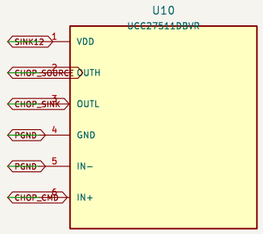
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

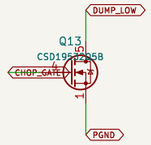
Sheet: /LVC Schmitt fault inputs without series diode offsets/
File: fault-interfaces.kicad_sch

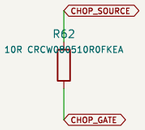
Title: LVC Schmitt fault inputs without series diode offsets

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 10/32

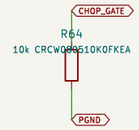
Autonomous OV path ignores bridge ENABLE. External resistor is required across BRAKE_DUMP_RESISTOR connector pins.

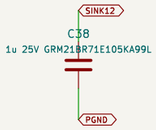
















Candidate: 2 x TE HSC3004R7J in series (9.4 ohm). Chassis/heatsink + NC thermal cutoff required.

No resistor cable clamp / fuse / SOA qualification yet. Loss of AUX supply loses bus sink: NOT FAIL SAFE.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

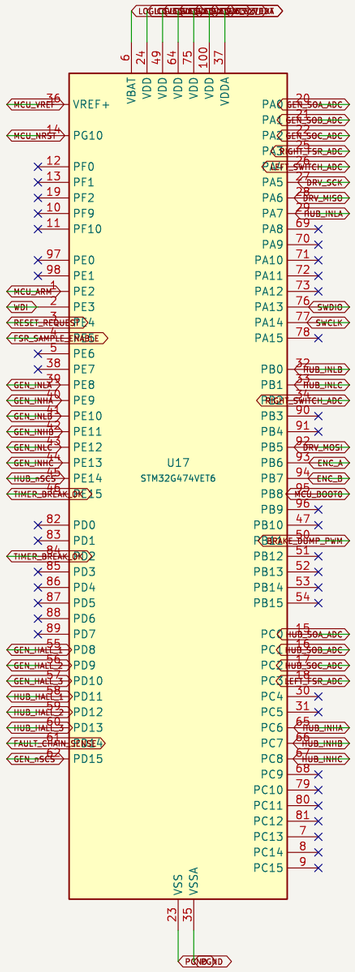
Sheet: /Independent brake chopper power stage/
File: brake-chopper.kicad_sch

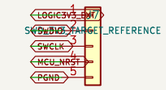
Title: Independent brake chopper power stage

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 11/32

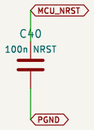
HSI16 internal clock / PLL option: no HSE or LSE fitted; PF0/PF1 and PC14/15 explicit NC.

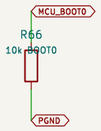
BOOT0 low; keep NRST function and configure boot option bytes before use. No firmware supplied.











BKIN active LOW required in TIM1/TIM8. Separate hardware ENABLE path remains intact.

NC GPIO: leave analog mode; HSI accuracy / PLL / ADC timing require validation.

MCU_ARM / RESET_REQUEST 10k and WDI 1k default-low resistors are on fault-latch page.

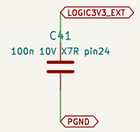
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

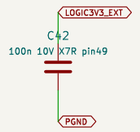
Sheet: /STM32G474VET6 LQFP100: power, reset, debug and control/
File: mcu-core.kicad_sch

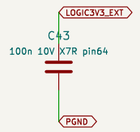
Title: STM32G474VET6 LQFP100: power, reset, debug and control

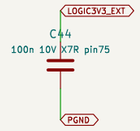
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 12/32

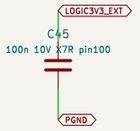
One 100n per VDD pin (24,49,64,75,100) plus VBAT pin6. Place each at pin / ground return.

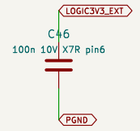


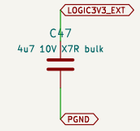


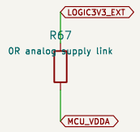


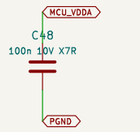


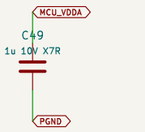


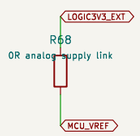


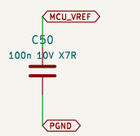


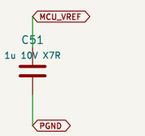












VREFBUF MUST remain disabled: external 3V3 reference. VDDA/VREF tied via OR; not precision isolated.



MCU_VDDA: arrives through R67 OR analog link, not a power-OUT

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

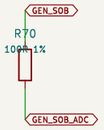
Sheet: /MCU 3V3 decoupling — external supervised rail/
File: mcu-power.kicad_sch

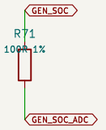
Title: MCU 3V3 decoupling — external supervised rail

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 13/32

100R / 1nF candidate filters: settling / sample aperture / CSA gain remain unqualified.

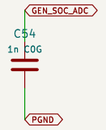




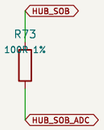


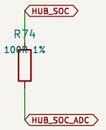




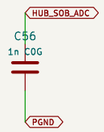


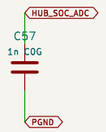


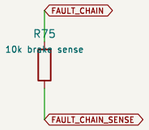


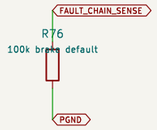


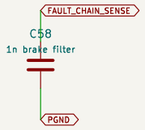












FAULT_CHAIN_SENSE is status ONLY. E-stop / thermal comparator and fault path do not depend on firmware.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Six CSA ADC input filters and brake status sense/
File: mcu-analog.kicad_sch

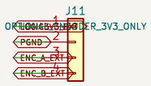
Title: Six CSA ADC input filters and brake status sense

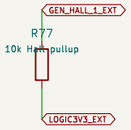
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 14/32

3V3 ONLY: open-collector Hall / encoder or compatible 3V3 outputs. NO 5V tolerance claim.



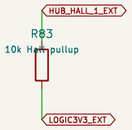






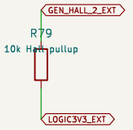












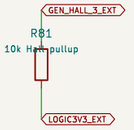


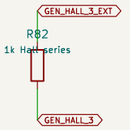


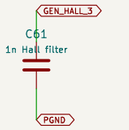


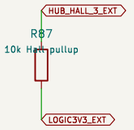






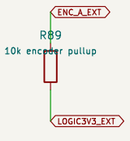




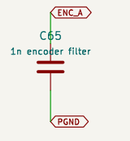


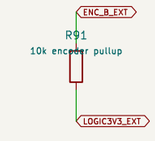




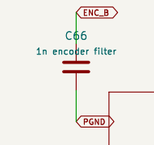












Encoder PB6/PB7: TIM4_CH1/CH2 AF2 DS12288r5 T13 p74. Cable ESD/EMC protection unresolved.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

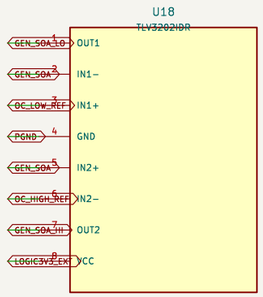
Sheet: /3V3 Hall interfaces and optional quadrature encoder/
File: mcu-sensors.kicad_sch

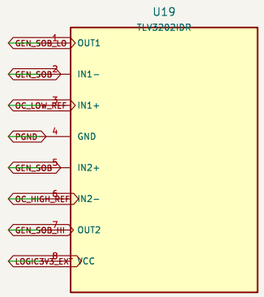
Title: 3V3 Hall interfaces and optional quadrature encoder

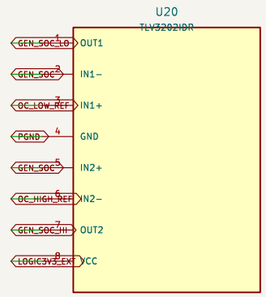
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 15/32

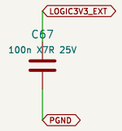
TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

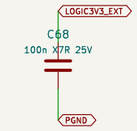
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

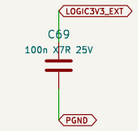


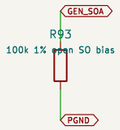


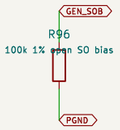


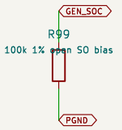


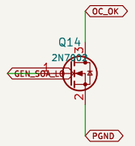


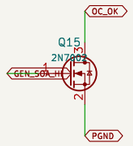


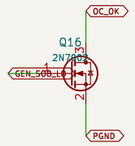


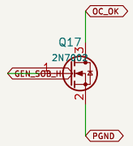


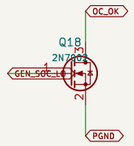


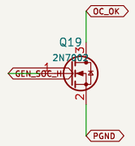


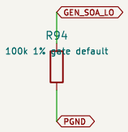


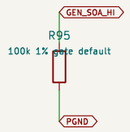


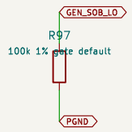


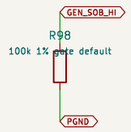


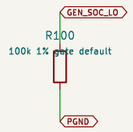


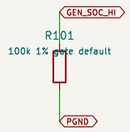












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

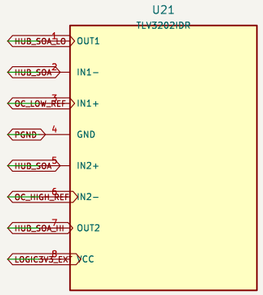
Sheet: /GEN independent raw CSA window trips/
File: gen−fast−oc.kicad_sch

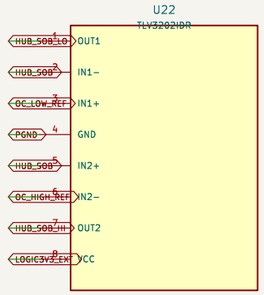
Title: GEN independent raw CSA window trips

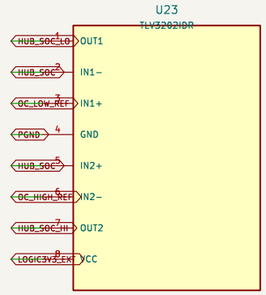
Size: A3	Date:	Rev: A−PARTIAL
KiCad E.D.A. 9.0.9		Id: 16/32

TLV3202IDR: positive SP−SN current DECREASES SO. Lower and upper limits both latch off.

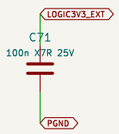
No PWM blanking added. Low−side blind intervals remain; DRV VDS OCP contract mandatory.

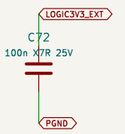


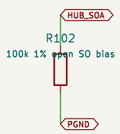


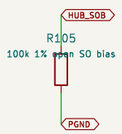


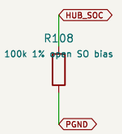


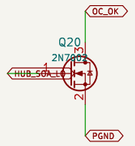


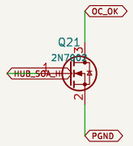


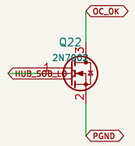


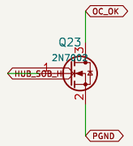


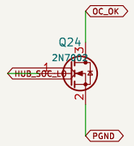


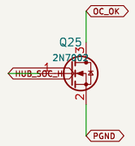


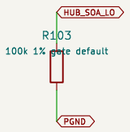


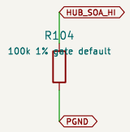


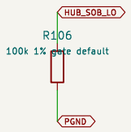


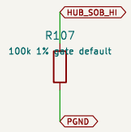


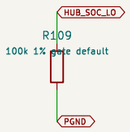


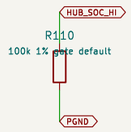












Push−pull comparators drive separate sink FETs; outputs are NOT shorted together.

Open SO pulls down; stuck midpoint / shunt open / power−loss safety NOT proven. UNSAFE DRAFT.

NOT SAFE TO ENERGIZE − SAFETY CIRCUITS UNFINISHED

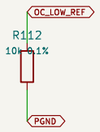
Sheet: /HUB independent raw CSA window trips/
File: hub−fast−oc.kicad_sch

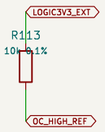
Title: HUB independent raw CSA window trips

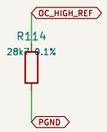
Size: A3	Date:	**Rev: A−PARTIAL**
KiCad E.D.A. 9.0.9		Id: 17/32

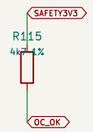
VREF tracks actual driver LOGIC3V3_EXT: 0.852713 / 2.447287 V nominal at 3.3 V.











OC_OK directly enters U14.6, clears U7/U8 and U16: both ENABLE and TIM1/TIM8 break.

No external hysteresis or blanking: first crossing latches. Noise/settling and sink delay unqualified.

Nominal trip +/−19.932 A; provisional measurement +/−30 A. Commissioning cap 10 A, NOT release.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /Ratiometric OC window and wired sink fault/
File: fast-oc-reference.kicad_sch

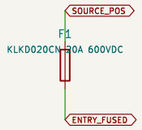
Title: Ratiometric OC window and wired sink fault

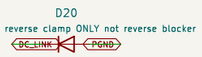
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 18/32

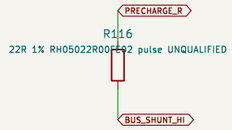
NOT SAFE TO ENERGIZE integrated drive. No automatic sequencing or reverse-blocking guarantee.

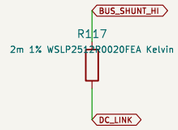
36-58.4V bench interface, 15A TARGET ONLY. Disconnect PM machines before source removal.

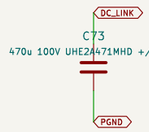


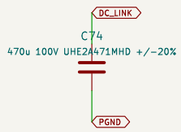


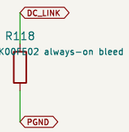


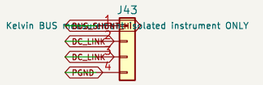






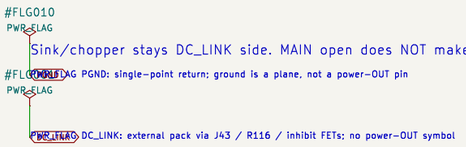






J41/J42 removed. MAIN/PRE physical pairs are inhibited on separate sheets. NO ENERGIZE.

J43 raw high common mode: NOT MCU ADC. D20/fuse reverse-fault coordination UNQUALIFIED.



NOT SAFE TO ENERGIZE - SAFETY CIRCUITS UNFINISHED

Sheet: /Bounded bench DC entry - active pairs inhibited/
File: dc-entry.kicad_sch

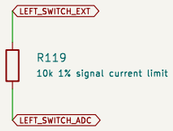
Title: Bounded bench DC entry - active pairs inhibited

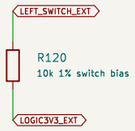
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 19/32

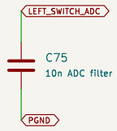
Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

J44
LEFT_BINARY_SWITCH_ONLY









D32
PESD3V3U1UL



D36
BAT54S

External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT supervised binary brake interface/
File: brake-intent-left.kicad_sch

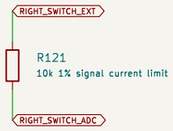
Title: LEFT supervised binary brake interface

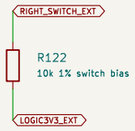
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 20/32

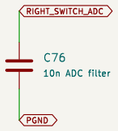
Candidate only: reversible non-load-bearing lever sensors. NO mechanical brake cable alteration.

J45
RIGHT_BINARY_SWITCH_ONLY





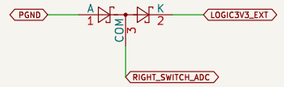




D33
PESD3V3U1UL



D37
BAT54S



External switch harness: 4k7 series + (10k parallel NC switch). Release=1.055V; press=1.964V.

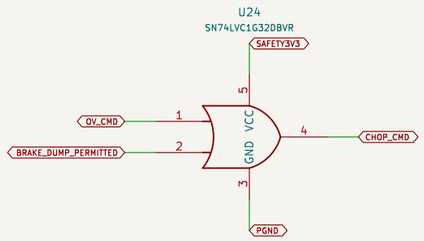
Open=3V3 / short=0V faults via ADC; firmware NOT implemented. Clamp injection/power-off NOT qualified.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

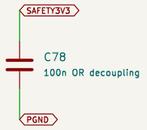
Sheet: /RIGHT supervised binary brake interface/
File: brake-intent-right.kicad_sch

Title: RIGHT supervised binary brake interface

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 21/32





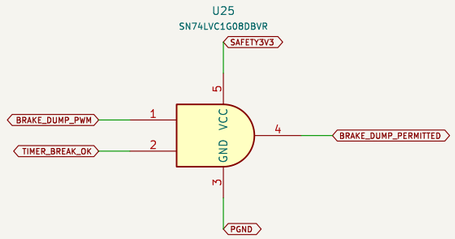


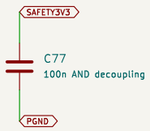
OV always overrides MCU PWM. Normal PWM must require valid rider request AND cool dump permission.

PWM AND TIMER_BREAK_OK: watchdog / E-stop / thermal fault kill normal dump. OV bypass retained.

No torque-sign hardware guard. Firmware must disarm on sensor fault; watchdog / OC / E-stop remain hardware kills.

Firmware entirely unimplemented. Do not energize; brake command control only a design contract.





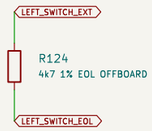
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

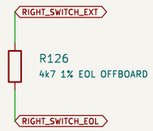
Sheet: /Rider dump PWM OR independent emergency overvoltage/
File: brake-dump-command.kicad_sch

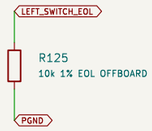
Title: Rider dump PWM OR independent emergency overvoltage

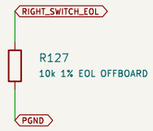
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 22/32

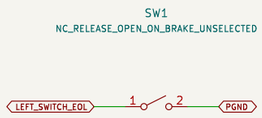
OFFBOARD parts must live at lever, not board. NC contact opens on rider braking. No brake cable modification.

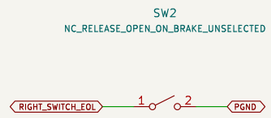












Switch symbol drawn open (braking). Model cables unplugged by removing BOTH EOL resistors and switch.

Welded contact or resistor short can imitate release; FSR plausibility is firmware only, not independent safety.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /OFFBOARD supervised switch end-of-line network – candidate/
File: brake-switch-harness.kicad_sch

Title: OFFBOARD supervised switch end-of-line network – candidate

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 23/32

Interlink 34-00018 external 402 short-tail pad; connector mate / mechanics UNQUALIFIED.

LOW-VOLTAGE BENCH ONLY. TVS / clamps candidates, NOT battery-short protection. No torque-sign guard.

J46
LEFT_FSR_34-00018_EXTERNAL

LEFT_FSR_EXC_CABLE
LEFT_FSR_RETURN_CABLE
1
2

MCU_VREF
R128
10k 1% FSR excitation
LEFT_FSR_EXC_CABLE

LEFT_FSR_RETURN_CABLE
R129
1k 1% FSR return
LEFT_FSR_SENSE

LEFT_FSR_SENSE
R130
10k 0.1% FSR measure
PGND

LEFT_FSR_SENSE
C79
100n FSR sense
PGND

LEFT_FSR_SENSE
R131
10k 1% OPA input limit
LEFT_FSR_OP_IN

MCU_VREF
C80
100n OPA bypass
PGND

U26
OPA333AIDBVR

LEFT_FSR_BUF	1	OUT
PGND	2	V-
LEFT_FSR_OP_IN	3	+IN
LEFT_FSR_BUF	4	-IN
MCU_VREF	5	V+

U27
TMUX1511PWR

FSR_SAMPLE_ENABLE	1	SEL1	D3	8	
LEFT_FSR_BUF	2	S1	S3	9	
LEFT_FSR_ISO_OUT	3	D1	SEL3	10	PGND
PGND	4	SEL2	D4	11	
	5	S2	S4	12	
	6	D2	SEL4	13	PGND
PGND	7	GND	VDD	14	LOGIC3V3_EXT

LEFT_FSR_ISO_OUT
R132
1k 1% ADC isolation
LEFT_FSR_ADC

LEFT_FSR_ADC
R133
10k 1% ADC bleed
PGND

LEFT_FSR_ADC
C81
100n ADC reservoir
PGND

FSR_SAMPLE_ENABLE
R134
100k enable default off
PGND

LOGIC3V3_EXT
C82
100n mux bypass
PGND

MCU_VREF = excitation / OPA supply; LOGIC3V3_EXT = mux supply. PGND common; no isolation of ground.

Enable PE5 pin4 default low; 20ms blanking + plausibility pending firmware / bench validation.

Binary switch authoritative by firmware CONTRACT ONLY. Invalid FSR cannot authorize positive torque.

#FLG006
PWR_FLAG

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT selected FSR low-voltage bench frontend/
File: fsr-left.kicad_sch

Title: LEFT selected FSR low-voltage bench frontend

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 24/32

#FLG007
PWR_FLAG
MCU_VREF: MCU internal reference output on U17 pin 36
MCU_VREF

Connector TVS: cathode1 cable, anode2 ground. BAT54S: anode1 ground, cathode2 rail, junction3 sense.

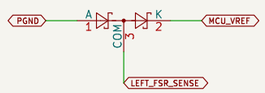
D38
PESD3V3U1UL



D39
PESD3V3U1UL



D40
BAT54S



NOT QUALIFIED: clamp voltage, leakage, negative pulses, rail sink / powered-off backfeed, ESD energy.

Battery short is a RELEASE BLOCKER. No external cable driver; segregated current-limited LV bench only.

TVS is NOT a 3.3V limiter. Secondary clamp can backfeed MCU_VREF. No vehicle energization authorization.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /LEFT FSR candidate ESD and secondary clamps – unqualified/
File: fsr-protection-left.kicad_sch

Title: LEFT FSR candidate ESD and secondary clamps – unqualified

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 25/32

Interlink 34-00018 external 402 short-tail pad; connector mate / mechanics UNQUALIFIED.

LOW-VOLTAGE BENCH ONLY. TVS / clamps candidates, NOT battery-short protection. No torque-sign guard.

J47
RIGHT_FSR_34-00018_EXTERNAL
RIGHT_FSR_EXC_CABLE
RIGHT_FSR_RETURN_CABLE
1
2

MCU_VREF
R135
10k 1% FSR excitation
RIGHT_FSR_EXC_CABLE

RIGHT_FSR_RETURN_CABLE
R136
1k 1% FSR return
RIGHT_FSR_SENSE

RIGHT_FSR_SENSE
R137
10k 0.1% FSR measure
PGND

RIGHT_FSR_SENSE
C83
100n FSR sense
PGND

RIGHT_FSR_SENSE
R138
10k 1% OPA input limit
RIGHT_FSR_OP_IN

MCU_VREF
C84
100n OPA bypass
PGND

U28
OPA333AIDBVR
RIGHT_FSR_BUF 1 OUT
PGND 2 V-
RIGHT_FSR_OP_IN 3 +IN
RIGHT_FSR_BUF 4 -IN
MCU_VREF 5 V+

U29
TMUX1511PWR
FSR_SAMPLE_ENABLE 1 SEL1
RIGHT_FSR_BUF 2 S1
RIGHT_FSR_ISO_OUT 3 D1
PGND 4 SEL2
5 S2
6 D2
PGND 7 GND
8 D3
9 S3
10 SEL3 PGND
11 D4
12 S4
13 SEL4 PGND
14 VDD LOGIC3V3_EXT

RIGHT_FSR_ISO_OUT
R139
1k 1% ADC isolation
RIGHT_FSR_ADC

RIGHT_FSR_ADC
R140
10k 1% ADC bleed
PGND

RIGHT_FSR_ADC
C85
100n ADC reservoir
PGND

FSR_SAMPLE_ENABLE
R141
100k enable default off
PGND

LOGIC3V3_EXT
C86
100n mux bypass
PGND

MCU_VREF = excitation / OPA supply; LOGIC3V3_EXT = mux supply. PGND common; no isolation of ground.

Enable PE5 pin4 default low; 20ms blanking + plausibility pending firmware / bench validation.

Binary switch authoritative by firmware CONTRACT ONLY. Invalid FSR cannot authorize positive torque.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /RIGHT selected FSR low-voltage bench frontend/
File: fsr-right.kicad_sch

Title: RIGHT selected FSR low-voltage bench frontend

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 26/32

Connector TVS: cathode1 cable, anode2 ground. BAT54S: anode1 ground, cathode2 rail, junction3 sense.

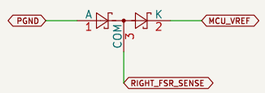
D41
PESD3V3U1UL



D42
PESD3V3U1UL



D43
BAT54S



NOT QUALIFIED: clamp voltage, leakage, negative pulses, rail sink / powered-off backfeed, ESD energy.

Battery short is a RELEASE BLOCKER. No external cable driver; segregated current-limited LV bench only.

TVS is NOT a 3.3V limiter. Secondary clamp can backfeed MCU_VREF. No vehicle energization authorization.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

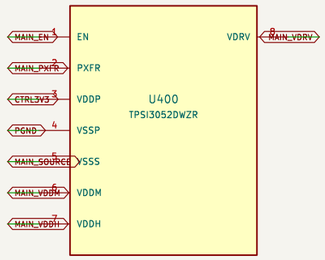
Sheet: /RIGHT FSR candidate ESD and secondary clamps – unqualified/
File: fsr-protection-right.kicad_sch

Title: RIGHT FSR candidate ESD and secondary clamps – unqualified

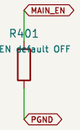
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 27/32

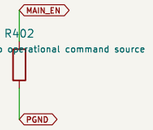
NO ENERGIZE: OR EN inhibit fitted; sequencer and CTRL3V3 buck NOT IMPLEMENTED.

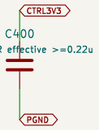
150V IRFP4568PbF TO-247AC G=1 D=2 S=3 (PD-96175). SOA, clamp and thermal qualification remain OPEN.



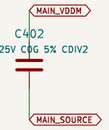


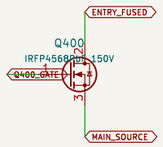


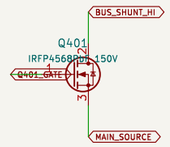


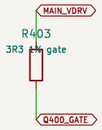


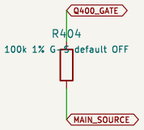


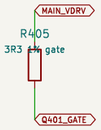


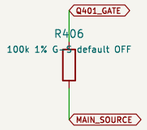






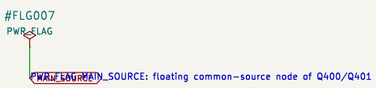






Common-source node floats: NEVER PGND. CDIV2/CDIV1 worst 470*.95/(100*1.05)=4.25 > 3.

EN not gate feedback. No MAIN_PROVEN, timeout, PM-quiet or current detector claimed.



MAIN_SOURCE: floating common-source node of Q400/Q401

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

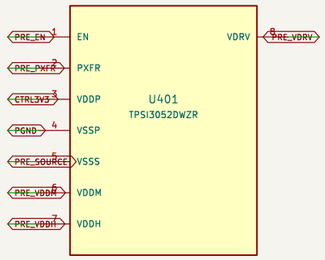
Sheet: /MAIN back-to-back candidate — hardware inhibited/
File: entry-main.kicad_sch

Title: MAIN back-to-back candidate — hardware inhibited

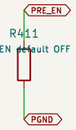
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 28/32

NO ENERGIZE: OR EN inhibit fitted; sequencer and CTRL3V3 buck NOT IMPLEMENTED.

150V IRFP4568PbF TO-247AC G=1 D=2 S=3 (PD-96175). SOA, clamp and thermal qualification remain OPEN.





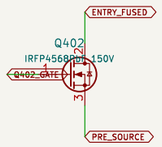


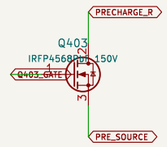


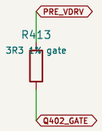


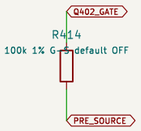


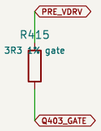


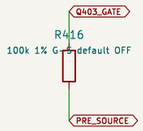






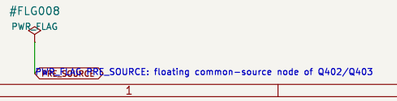






Common-source node floats: NEVER PGND. CDIV2/CDIV1 worst 470*.95/(100*1.05)=4.25 > 3.

EN not gate feedback. No MAIN_PROVEN, timeout, PM-quiet or current detector claimed.



NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /PRE back-to-back candidate — hardware inhibited/
File: entry-pre.kicad_sch

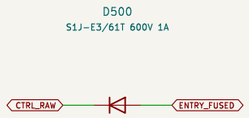
Title: PRE back-to-back candidate — hardware inhibited

Size: A3	Date:	**Rev: A-PARTIAL**
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36–58.4V pack –> ENTRY_FUSED –> D500 –> CTRL_RAW. Upstream MAIN; no bus dependency.

TI LM5164 SNVSAU4D adapted Type–3 network. 0.8W PACK INPUT acceptance cap, NOT output rating.

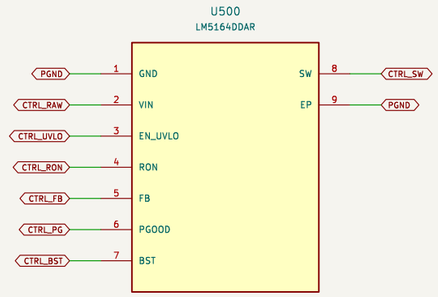
NOT SAFE TO ENERGIZE. No PGOOD interlock or whole–system reverse/OV protection claimed.







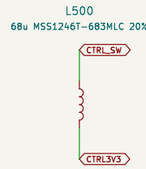


























PGOOD to CTRL_PG: rail–partition interlock; no MAIN proof claimed. U400/U401 EN OR inhibits remain fitted.

200V input ceramics: effective C / surge / damping OPEN; 2225 footprint candidate, layout not qualified.

#FLG002
PWR_FLAG

D500 protects this positive supply branch only; no full–system reverse–polarity or transient proof.

#FLG003
PWR_FLAG

CTRL_RAW: DC_LINK through D500 S1J rectifier – passive, not a driver

CTRL3V3: U500 LM5164 buck output via L500; SW is not a power–OUT

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /CTRL3V3 pack–side buck – unqualified, enables inhibited/
File: ctrl–supply.kicad_sch

Title: CTRL3V3 pack–side buck – unqualified, enables inhibited

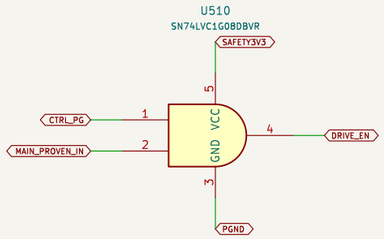
Size: A3	Date:	Rev: A–PARTIAL
KiCad E.D.A. 9.0.9		Id: 30/32

NOT SAFE TO ENERGIZE. Partition of the former single AUX12V; no release or energization claim.

SINK12 = U3 R-78HB12-0.5 output: always bus-fed sink/safety cluster. 1.1 W input envelope, pre-proof only.

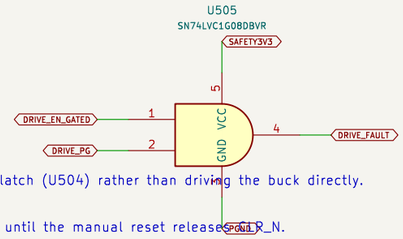
DRIVE12 = bus LM5164 (TI 12V reference values). LOGIC3V3_EXT = AP63203 from DRIVE12. Both EN held low by 100k.

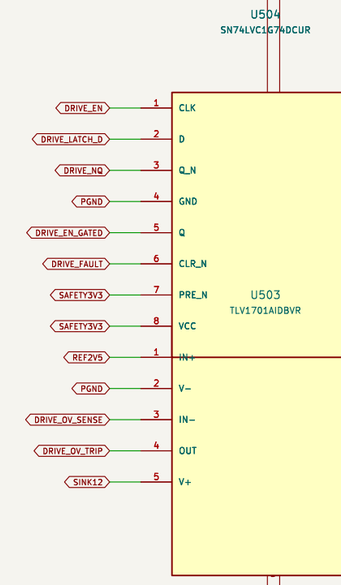
DRIVE_EN = CTRL_PG AND MAIN_PROVEN_IN. MAIN_PROVEN_IN has a FITTED OR inhibit to PGND.

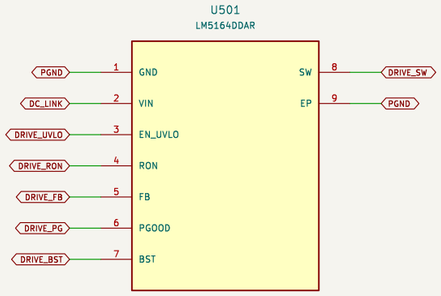


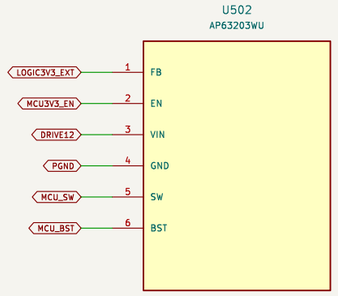
U510 is powered from SAFETY3V3, the always-on safety rail. Losing it floats Y: 100k pulldowns keep both EN low. Its output is DRIVE_EN, which now clocks the fault latch (U504) rather than driving the buck directly.

Enable chain: DRIVE_EN -> U504 clock -> Q = DRIVE_EN_GATED -> U501 EN/UVLO via R514. A latched fault holds DRIVE_EN_GATED low, so the buck stays disabled until the manual reset releases CLR_N.









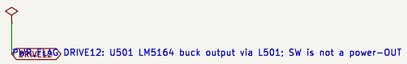
U501 is TI figure 7-1 (12 V/1 A/300 kHz) with RRON=100k and RFB1/RFB2=453k/49.9k. PGOOD drives DRIVE_PG only.

U502 pin map per Diodes TSOT26 table; FB tied to VOUT for the fixed 3.3 V variant.

U503 output asserts DRIVE_OV_TRIP low when DRIVE12 exceeds 2.5V*(R523+R524)/R524 = 16.25 V nominal. Independent of the LM5164 FB node, so an open FB divider cannot hide the fault. It drives the latch (U504), not the buck directly: see RAIL-16.

U505 gives DRIVE_PG its consumer: any PGOOD-low clears the latch, so a buck that drops out of regulation cannot silently keep the enable asserted.

U504 is the DRIVE12 fault latch: Q holds the enable off after one OVP trip until CLR_N is released. Powered from SAFETY3V3 so it survives a DRIVE12 collapse; PRE_N tied high so it cannot be set remotely.



NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

Sheet: /Rail partition: SINK12 always-on, DRIVE12/MCU3V3 delayed and inhibited/
File: rail-partition.kicad_sch

Title: Rail partition: SINK12 always-on, DRIVE12/MCU3V3 delayed and inhibit

Size: A3	Date:	**Rev: A-PARTIAL**
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L501
68u SPM10065VC-680M-D 20% AEC-Q200
DRIVE_SW
DRIVE12

L502
3.9u 2A
MCU_SW
LOGIC3V3_EXT

DC_LINK
C510
2.2u 200V X7R 22252C225KAT2A
PGND

DC_LINK
C511
2.2u 200V X7R 22252C225KAT2A
PGND

DRIVE_BST
C512
2.2n 50V GCM155R71H222KA37D
DRIVE_SW

DRIVE12
C513
22u 25V TMK325B7226KMHT
PGND

DRIVE12
C514
22u 25V TMK325B7226KMHT
PGND

DRIVE_RAMP
C515
3.3n 100V CGA3E2X7R2A332K080AA
DRIVE12

DRIVE_RAMP
C516
56p 50V C0G
DRIVE_FB

DRIVE12
C517
10u 25V X7R
PGND

LOGIC3V3_EXT
C518
22u 16V X7R
PGND

LOGIC3V3_EXT
C519
22u 16V X7R
PGND

MCU_BST
C520
100n 25V X7R
MCU_SW

SAFETY3V3
C521
100n 25V X7R
PGND

DRIVE_RON
R510
100k 1% 0.1W TI reference
PGND

DRIVE12
R511
453k 1% 0.1W TI reference
DRIVE_FB

DRIVE_FB
R512
49.9k 1% RG1608P-4992-B-T5
PGND

DRIVE_SW
R513
453k 1% 0.1W TI reference
DRIVE_RAMP

DRIVE_EN_GATED
R514
82k 1% 0.1W >=100V
DRIVE_UVLO

DRIVE_UVLO
R515
100k 1% DELAY EN pulldown
PGND

DRIVE_EN
R516
47k 1% EN series
MCU3V3_EN

MCU3V3_EN
R517
100k 1% DELAY EN pulldown
PGND

MAIN_PROVEN_IN
R518
100k 1% MAIN_PROVEN default low
PGND

MAIN_PROVEN_IN
R519
0R FIXED INHIBIT — no MAIN_PROVEN detector fitted
PGND

CTRL3V3
R520
10k CTRL PGOOD pullup
CTRL_PG

SINK12
R521
47k 1% DRIVE_PG pullup (PGOOD abs max 14V)
DRIVE_PG

DRIVE12
R523
649k 0.1% DRIVE12 OVP upper leg
DRIVE_OV_SENSE

DRIVE_OV_SENSE
R524
118k 0.1% DRIVE12 OVP lower leg
PGND

DRIVE_OV_SENSE
C523
100p 50V C0G OVP sense filter
PGND

DRIVE_LATCH_D
R525
100k 1% DRIVE latch D tied low
PGND

SAFETY3V3
R526
47k 1% DRIVE_FAULT pullup
DRIVE_FAULT

DRIVE_FAULT
R527
100k 1% manual reset series (SW to PGND)
DRIVE_RESET_SW

DRIVE_NQ
R528
10k 1% DRIVE_NQ load, keeps Q_N defined
PGND

D520
SS14 40V 1A OVP trip wired-OR
DRIVE_OV_TRIP
DRIVE_FAULT

D521
SS14 40V 1A PGOOD wired-OR
DRIVE_PG
DRIVE_FAULT

D522
SS14 40V 1A manual reset wired-OR
DRIVE_RESET_SW
DRIVE_FAULT

J48
Conn_01x02 reset header, SW to PGND
DRIVE_RESET_SW
1
2
PGND

R519 OR keeps MAIN_PROVEN_IN low, so DRIVE_EN stays low: both delayed rails are OFF by construction.

U501 PGOOD drives DRIVE_PG only: the always-on ANALOG_OK channel is no longer gated by a delayed rail.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Rail partition passive networks — unqualified/
File: rail-passives.kicad_sch

Title: Rail partition passive networks — unqualified

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 32/32